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--- Load Data ---

```
file = 'linklite_data_4.csv';
data = readmatrix(file);           % 20x9 numeric matrix
hdr  = readcell(file, 'Range', '1:1'); % header row (cell array of
strings)
```

```
brightness = 1;      % (%)
scaled_duty = 3;      % (%)
measured_duty = 4;    % (%)
```

```
total_current = 5;      % mA
power_consumption = 6; % W
voltage_drop = 7;       % V
voltage_rail = 8;       % V
led_current = 9;        % mA
```

```
efficiency = ((data(:,led_current) * 28) .* data(:,voltage_rail)) ./ ...
              ((data(:,total_current) - 72) * 15); % (%)
```

```
buck_output = efficiency .* data(:,total_current) .* 15 ./
data(:,voltage_rail)
```

```
buck_output =
```

```
1.0e+03 *
```

```
NaN
NaN
0.0818
0.1194
0.1494
0.1675
0.1875
0.2215
0.2546
0.2960
0.3427
0.3925
0.4502
0.5219
0.5954
0.6698
1.1706
```

```
1.3122
1.4525
1.5970
1.7941
```

graphing data

```
figure;

sgtitle("Linklite Power Metrics", 'fontsize', 44);

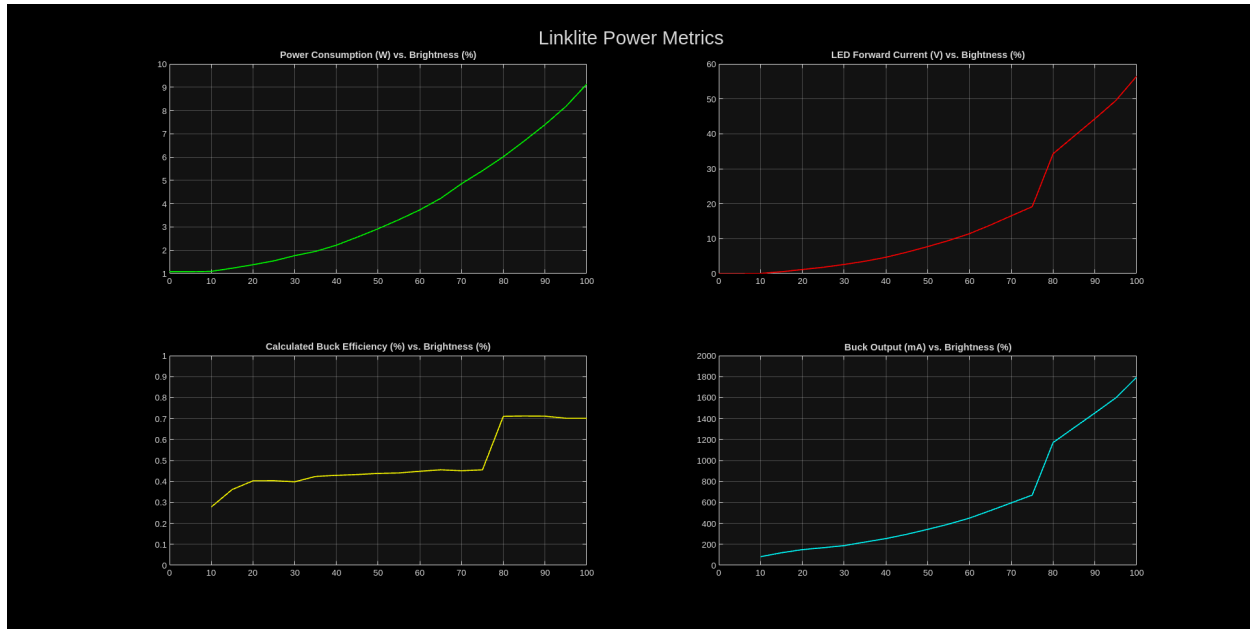
subplot(2,2,1);
plot(data(:,brightness), data(:,power_consumption), '-c', 'LineWidth', 2.5,
'MarkerSize', 10, 'Color', 'g');
title("Power Consumption (W) vs. Brightness (%)", 'FontSize', 28)
xlim([0 100]);
grid on;

subplot(2,2,2);
plot(data(:,brightness), data(:,led_current), 'LineWidth', 2.5, 'Color', 'r');
xlim([0 100]);
ylim([0 60]);
title("LED Forward Current (V) vs. Bightness (%)", 'FontSize', 28);
grid on;

subplot(2,2,3);
plot(data(:,brightness), efficiency, 'LineWidth', 2.5, 'Color', 'y');
xlim([0 100]);
ylim([0 1]);
title("Calculated Buck Efficiency (%) vs. Brightness (%)", 'FontSize', 28);
grid on;

subplot(2,2,4);
plot(data(:,brightness), buck_output, 'LineWidth', 2.5, 'color', 'c');
xlim([0 100]);
ylim([0 2000]);
title("Buck Output (mA) vs. Brightness (%)", 'FontSize', 28);
grid on;

for k = 1:4
    subplot(2,2,k);
    % ... your plot, title, labels ...
    set(gca, 'FontSize', 20);
end
```



--- Load and Graph Experimental Data ---

figure

hold on

```
temp_file = 'files/data_006.csv';
temp_data = readmatrix(temp_file);
plot(temp_data(:,1),temp_data(:,2), 'LineWidth',2);

temp_file = 'files/data_010.csv';
temp_data = readmatrix(temp_file);
plot(temp_data(:,1),temp_data(:,2), 'LineWidth',2);

temp_file = 'files/data_008.csv';
temp_data = readmatrix(temp_file);
plot(temp_data(:,1),temp_data(:,2), 'LineWidth',2);

temp_file = 'files/data_009.csv';
temp_data = readmatrix(temp_file);
plot(temp_data(:,1),temp_data(:,2), 'LineWidth',2);

temp_data_1 = readmatrix('files/data_011.csv');
temp_data_2 = readmatrix('files/data_012.csv');
temp_data_3 = readmatrix('files/data_013.csv');

% shortest row count across the three
n = min([size(temp_data_1,1), size(temp_data_2,1), size(temp_data_3,1)]);

% trim each to n rows
d1 = temp_data_1(1:n, :);
d2 = temp_data_2(1:n, :);
```

```
d3 = temp_data_3(1:n, :);

averaged_temp = (d1(:,1) + d2(:,1) + d3(:,1)) ./ 3;
plot(averaged_temp, d1(:,2));

temp_file = 'files/data_014.csv';
temp_data = readmatrix(temp_file);
plot(temp_data(:,1), temp_data(:,2), 'Linewidth', 2);

hold off
grid on

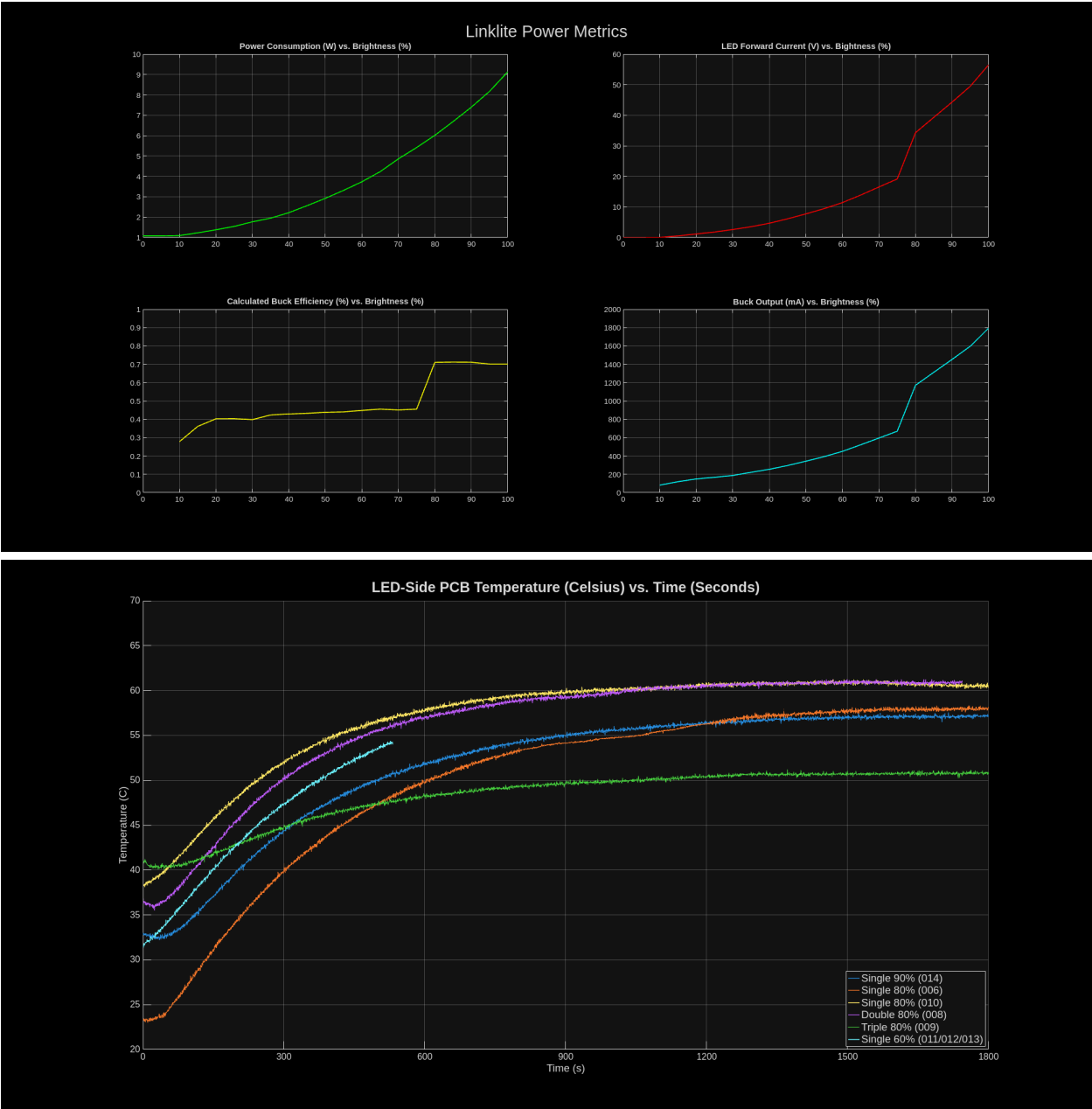
set(gca, 'FontSize', 24);

xticks(0:300:1800);
xlim([0 1800]);

yticks(20:5:70);
ylim([20 70]);

xlabel("Time (s)", FontSize=28);
ylabel("Temperature (C)", FontSize=28);
title("LED-Side PCB Temperature (Celsius) vs. Time (Seconds)", FontSize=38);

legend( ...
    'Single 90% (014)', ...
    'Single 80% (006)', ...
    'Single 80% (010)', ...
    'Double 80% (008)', ...
    'Triple 80% (009)', ...
    'Single 60% (011/012/013)', ...
    'fontsize', 28, ...
    'Location', 'Southeast' ...
);
```



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